



Basic examples of functional isometrics

Squat:

Rack stops placed halfway through the movement.

Perform a maximum isometric contraction for six seconds.

The contraction should be preceded by 3-4 dynamic repetitions.

Flat bench:

Stops positioned at the top of the movement.

Maximum contraction for 4-5 seconds.

The contraction should be preceded by 3-4 dynamic repetitions.

Block pulls:

Stopper positioned just above the knee.

Isometric contraction for 4-6 seconds.

The contraction should be preceded by 2-3 dynamic repetitions.

Practical applications

Functional isometrics can be used in a variety of contexts.

Advanced athletes: Ideal for improving near-maximal strength and

neuromuscular proficiency.

Post-injury recovery:

Strengthening specific areas without loading the entire ROM. For example: can be used to rehabilitate a shoulder, knee, or any joint by working on limited portions of the movement.

Sport-specific training: Sports such as weightlifting, athletics, or team sports can benefit from improvements in explosive force in crucial positions.

Periodisation: They can be included as specific training blocks to break plateaus or prepare for competitions.

Comparison with other methods

Isometric training: Focuses on developing static strength. Unlike functional isometrics, it does not include a dynamic component.

Dynamic training: Places force across the entire range of motion and does not specifically target sticking points or deadlocks, which functional isometrics are designed to address.

Progressive overload: Involves the gradual increase of load over time. It

Core matter: Functional isometrics are based on the use of isometric contractions (where the muscle generates tension without changing length) integrated with dynamic movements. GETTYIMAGES

does not place particular emphasis on isometric contractions, which are central to functional isometrics.

Plyometrics: Develop explosive force and speed, but they do not concentrate on specific ranges of motion as functional isometrics do.

Functional isometrics do not constitute a 'revolution' in sports training. However, they have certain interesting characteristics compared to other methods. They represent an advanced strategy for athletes and S&C coaches who want to optimise strength and precision of movement. Structured application requires knowledge, planning, and careful supervision to minimise risks. Used correctly, they can offer significant benefits in terms of performance, injury prevention, and rehabilitation.